

THE GROTTO

Comprehensive Digital Ecosystem for Modern Bouldering

Technical Specification & Committee Proposal | Version 1.1.0



[IMAGE: Splash Screen]
Capture the Beta v1.1 text
with the blue pulse effect.

Figure 1: The Grotto high-performance loading sequence.

Project Mission

To provide a unified, mobile-first platform that enhances the climber's journey from discovery to completion while empowering route-setters with professional-grade management tools.

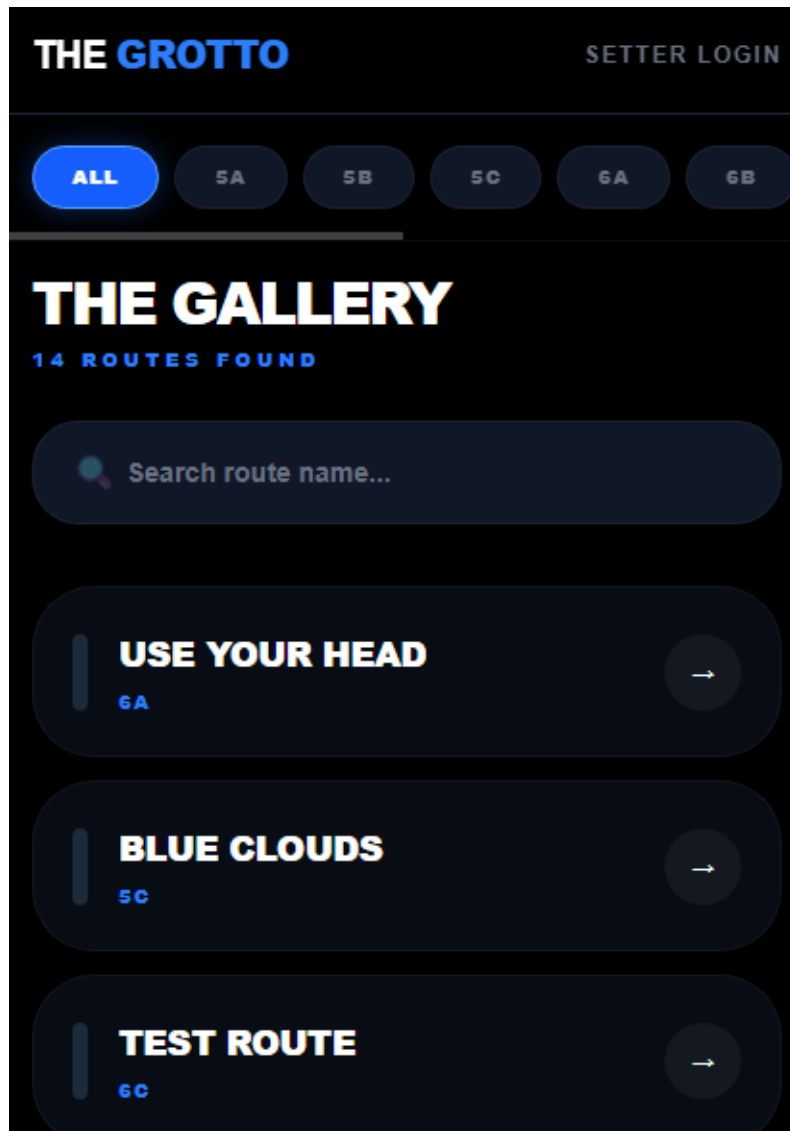
1 The Discovery Phase: Route Gallery & Filtering

The entry point of the application is designed to mimic modern social media feeds, making it instantly intuitive for all age groups.

1. Granular Grade Filtering

Climbers can filter the entire gym's inventory by difficulty (5a through 7c) or specific categories like "Newest" or "Sent."

- **Logic:** This prevents newer climbers from feeling overwhelmed and allows advanced climbers to jump straight to their projects.



[IMAGE: Main Gallery]
Screenshot of the home screen showing the horizontal grade filter bar and the route cards.

Figure 2: The dynamic filtering system allows for instant route discovery.

2 The Execution Phase: The Route Viewer

Once a route is selected, the application enters a high-performance viewing mode.

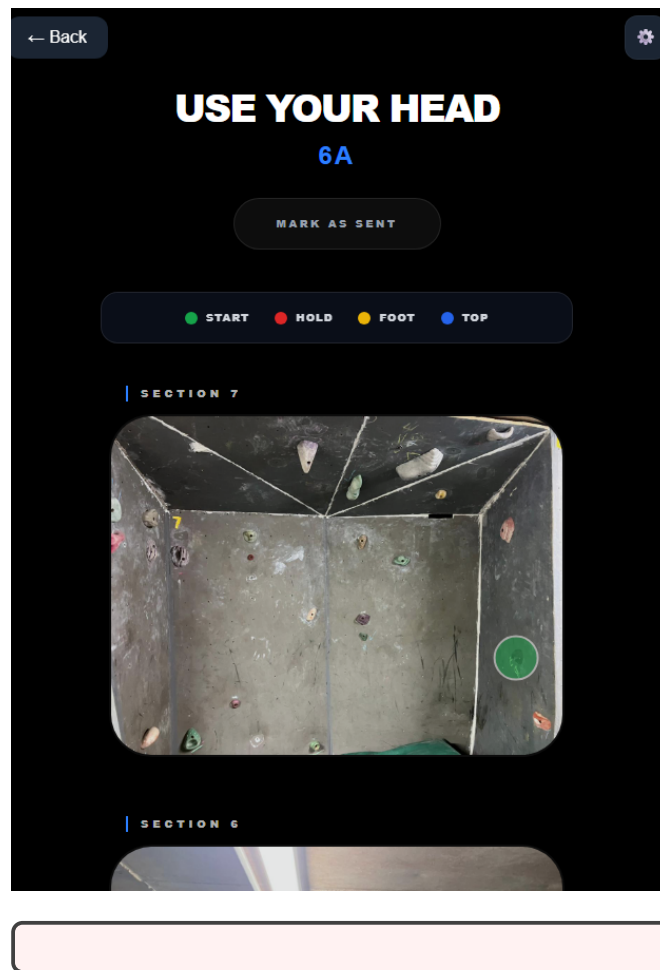
1. Multi-Wall Pathfinding

Traditional apps show one photo. The Grotto uses a **Linked Wall Engine**. If a route starts on the slab and finishes on the overhang, the app automatically stitches those specific images together.

2. Interactive Legend

- **Start:** Identifies mandatory starting holds.

- **Hold:** General hand/foot usage.
- **Foot:** Technical “foot-only” jibs that define the difficulty.
- **Top:** High-visibility finish indicator.



Route Viewer.

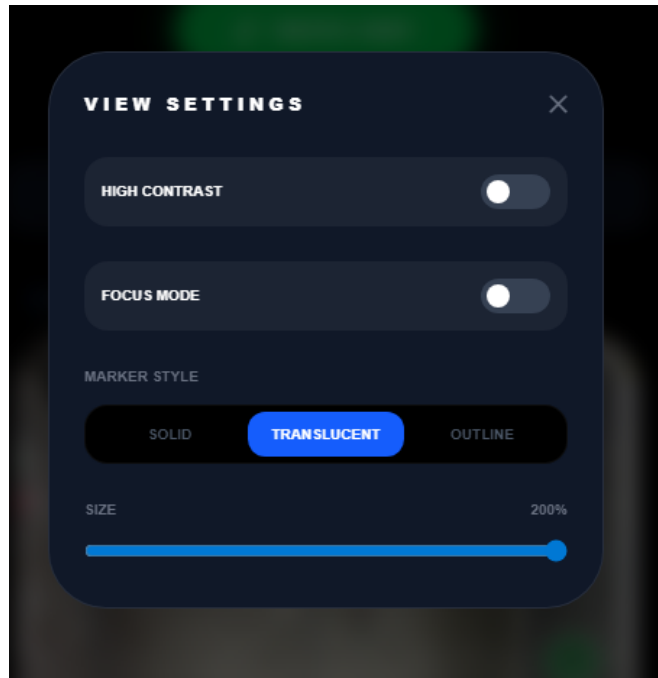
3 User Personalization: The Settings Suite

A key differentiator for The Grotto is its commitment to accessibility and user-defined comfort.

1. Adaptive Visual Preferences

Through the dedicated **Settings Tab**, users can modify the interface to suit their personal needs or the specific lighting conditions of the gym.

- **High Contrast Mode:** Thickens marker borders and increases saturation for better visibility on textured walls.
- **Marker Scaling:** Users can adjust the physical size of markers (e.g., $0.8x$ to $1.5x$) to ensure they do not obscure small holds or “footsies.”
- **Style Toggles:** Selection between “Solid,” “Translucent,” or “Outline” marker styles.



[IMAGE: Settings Menu]
Screenshot showing the sliders for
Marker Size and the toggles for
High Contrast / Marker Styles.

Figure 3: The customization engine ensuring accessibility for all climbers.

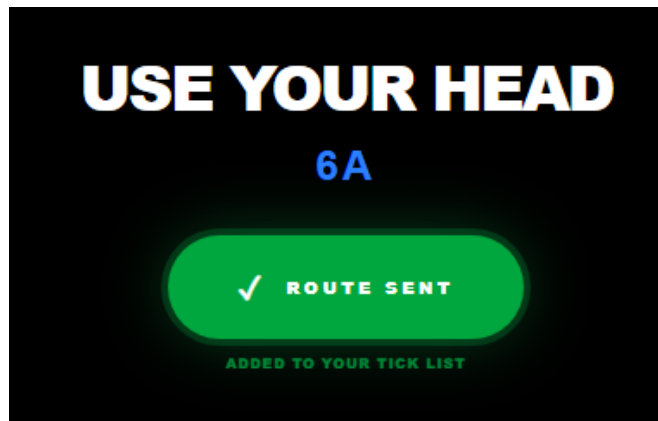
4 The Community Phase: Engagement & Progression

1. The "Sent It" Achievement

Marking a route as "Sent" is more than a checkbox. It is an achievement that triggers **haptic vibration feedback**, providing a physical sense of accomplishment.

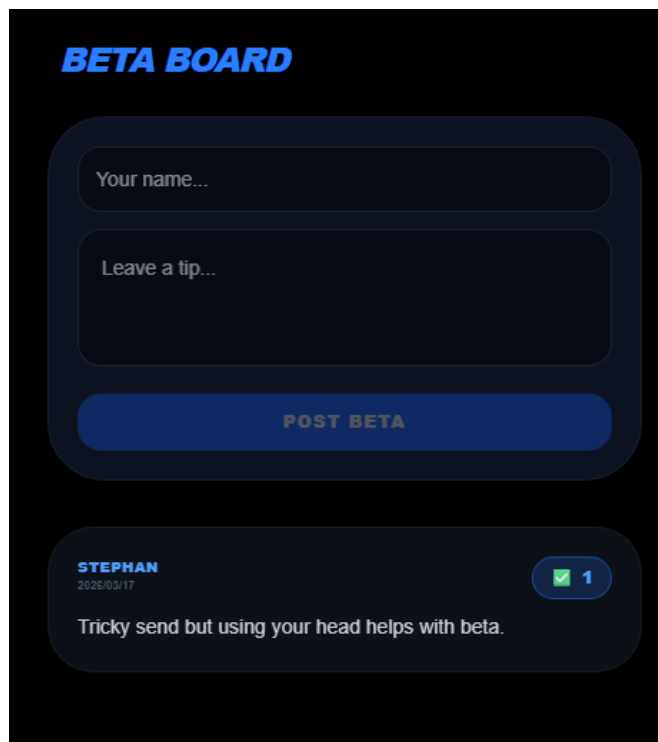
2. Social Beta & Feedback

The comment section allows for a live dialogue between the community and the setters.



[IMAGE: Achievement UI]
Screenshot of the "ROUTE SENT" green state and the user comments below it.

Figure 4: Closing the feedback loop between the climber and the community.



[IMAGE: Achievement UI]
Screenshot of the "BETA BOARD" user comments below it.

Figure 5: Closing the feedback loop between the climber and the community.

5 The Management Suite: Setter Operations

Behind a secure login, the app transforms into a professional gym management tool.

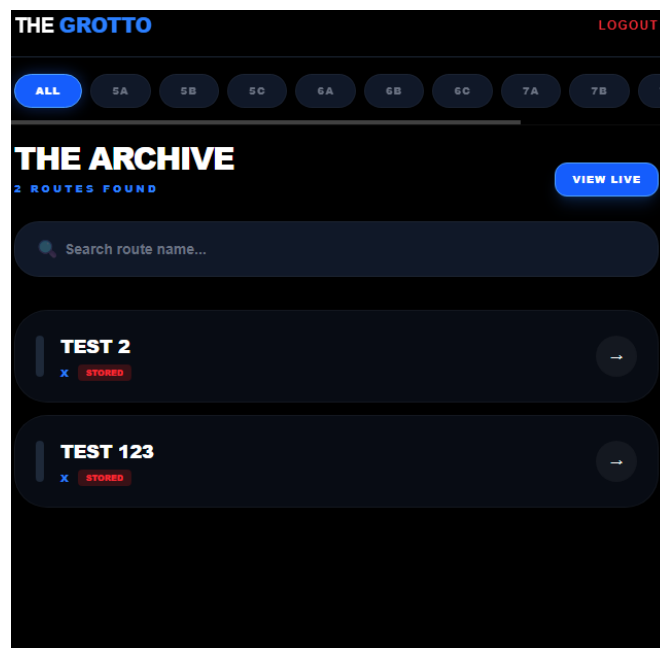
1. Lifecycle Management (The Archive Function)

Gyms must manage high turnover. The **Archive Function** is a critical administrative tool.

- **Instant Stripping:** When a wall is reset, setters simply toggle the “Archive” status. The route is instantly hidden from the public gallery.
- **Data Preservation:** Unlike deletion, archiving keeps the data accessible for internal reviews. The committee can see exactly how many 6a’s were set in a year or which setter had the most “Sent” routes.

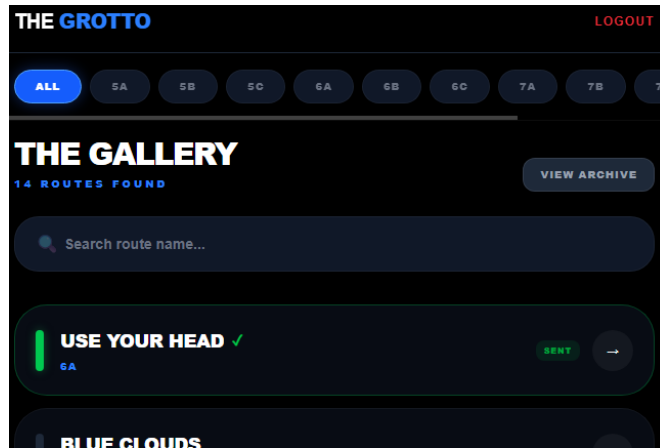
2. Active Route Editing

Setters can adjust marker positions or change a hold from a “Hold” to a “Foot” in real-time without needing to re-upload images.



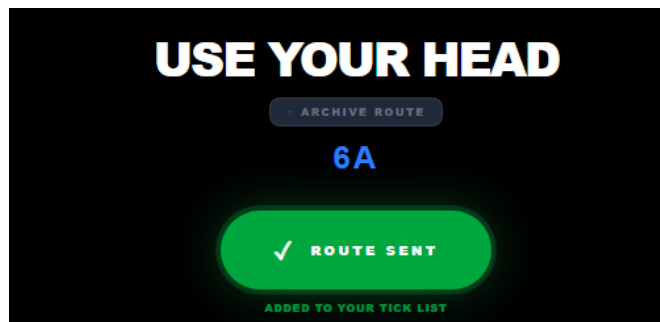
[IMAGE: Admin Archive Toggle]
Screenshot showing the Setter interface with the "The Archive").

Figure 6: The Archive system allows for clean, organized gym turnover management.



[IMAGE: Admin Archive Toggle]
Screenshot showing the Setter interface with the "View Archive " button.

Figure 7: The Archive system allows for clean, organized gym turnover management.



[IMAGE: Admin Archive Toggle]
Screenshot showing the Setter interface with the "Archive Route" button).

Figure 8: The Archive system allows for clean, organized gym turnover management.

6 Technical Superiority

- **Persistence:** Using `localStorage`, user progress and settings are saved even if the browser is closed.
- **Zero Install:** No App Store barriers. Users scan a QR code and the app is ready in seconds.
- **Performance:** Built with `Next.js` and `Supabase` for industrial-grade speed.

7 Conclusion

The Grotto is a complete transformation of the climbing gym experience. It brings professional-grade technology to the wall, making our gym more modern, organized, and community-focused.